

Farid Bagheri

AI Engineer & Researcher

PhD-level AI Researcher specializing in enhancing Large Language Models (LLMs) by grounding them in structured knowledge. My research focuses on architecting hybrid systems that integrate LLMs with Knowledge Graphs (KGs) to improve factuality, provide auditable reasoning, and boost performance in complex information retrieval tasks. I have extensive hands-on experience developing KG-LLM integrations, LLM-agent pipelines, and standalone Retrieval-Augmented Generation (RAG) systems. My background covers the full data lifecycle from constructing large-scale corpora via web scraping to advanced topic modeling with publications demonstrating that these fused systems significantly outperform traditional NLP baselines.

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Experience

Mercor	Remote
AI and prompt engineer	2025 - Present
• Evaluate and compare LLM outputs for accuracy, relevance, coherence, and instruction compliance. • Identify errors, hallucinations, and gaps between expected and generated responses. • Improve prompts and provide structured feedback to enhance model performance.	
Sorbonne Paris Nord University (LIPN)	Paris
Research Associate	Nov 2023 – Present
• Lead research on interpretable reviewer matching using LLMs and Knowledge Graphs. • Build modular LLM and agent pipelines with multi-model and judge-based selection. • Evaluate hybrid similarity methods on real-world academic datasets. • Produce peer-reviewed publications and international presentations.	

Sorbonne Paris Nord University	Paris
Graduate Lecturer	Nov 2024 – Present
Teaching Natural Language Processing and Knowledge Graphs, Machine Learning, and Deep Learning courses	

Sanem Plastik Company	Turkey
Quality Control Engineer and Data Analyst	2019– 2022
• Analyzed production-line quality data to identify performance issues and Estimated production costs and potential savings. • Collaborated with management to improve manufacturing and design standards.	

Previous experience:

- Industrial Engineer - Saba Palaye Factory (Persia, 2011– 2013)
- Marketing Manager - Raham Company (Persia, 2009– 2011)

Publications

LLM-Assisted Reviewer Assignment via Auditable Expertise Matching 2026

IEEE Access

Bagheri, F., Buscaldi, D., & Reforgiato Recupero, D.
<https://doi.org/10.1109/ACCESS.2026.3724622>

Reviewer Matching with LLMs: A Keyword-Driven, Interpretable Framework 2026

International Journal on Digital Libraries

Bagheri, F., Buscaldi, D., & Reforgiato Recupero, D.

Leveraging Knowledge Graphs and LLMs for Content-Based Reviewer Assignment 2025

Intelligent Information Systems

Bagheri, F., Buscaldi, D., & Reforgiato Recupero, D.
<https://doi.org/10.1007/s10844-025-01004-9>

Leveraging LLMs for Content-Based Reviewer Assignment in Computer Science Conferences 2025

GOLE–EGC Workshop

Bagheri, F., Buscaldi, D., & Reforgiato Recupero, D.

A Study on Content-Based Reviewer Assignment in the Semantic Web and Computer Science Domains 2024

Computación y Sistemas

Bagheri, F., Buscaldi, D., & Reforgiato Recupero, D.
<https://doi.org/10.13053/cys-28-4-5299>

Leveraging Return Prediction Approaches for Improved Value-at-Risk Estimation 2023

Data (MDPI)

Bagheri, F., Reforgiato Recupero, D., & Sirnes, E.
<https://doi.org/10.3390/data8080133>

Skills

Technical Skills

Python: Advanced

LLMs, KG–LLM, RAG, LLM agents, LangChain, Fine-tuning (LoRA/PEFT), PyTorch, scikit-learn, Hugging Face, NLTK, Flask/FastAPI, Ollama/OpenRouter



Additional Technical Skills

Databases: SQL, Time Series: Forecasting (ARIMA and walk-forward validation), Web: HTML - CSS



Soft Skills

Time Management, Teamwork, Documentation, Problem-solving



Awards

Ph.D. Fellowship in Mathematics and Computer Science
University of Cagliari -2022
A competitive full funding awarded for three years of Ph.D. candidacy based on academic merit and research potential

Education

University of Cagliari 2022 -Feb 2026
PhD in Mathematics and Computer Science
Research Focus: Grounding LLMs in Structured Data, Knowledge Graphs, NLP, Recommender Systems

Thesis: Semantic Reviewer Matching at Conference Scale with LLMGenerated Keywords and KnowledgeGraph Profiles.
<https://iris.unica.it/handle/11584/477586>

Dokuz Eylül University 2019 - 2022
M.Sc. in Industrial Engineering
Thesis: Evaluating the Successfulness of Start-up Businesses by Data Mining and Machine leraning Techniques

Azad University 2009 - 2014
B.Sc. in Industrial Engineering
Graduation Project: Marketing Strategy in Social Media Websites

Languages

English



Turkish



French



Interests

Electric Guitar, Snowboarding, Nature